

LEAD BRICKS

Fact Sheet

Lead Shielding

Metallic lead in the form of bricks, weights, plates, or rolls is used as movable radiation shielding in radiation areas, radioactive laboratories, and for spill containment of [positron emitters](#) and gamma radiation.

Exposure Hazards

Lead is known for its resistance to corrosion, but lead oxides, lead sulfates, and lead carbonates (i.e., lead salts) can form on the surface of lead bricks, due to oxidation from exposure to air, water, and humid environments. Lead salts (see image) are visible as a powdery coating that can appear white, yellow, or other colors. They generally form on metallic lead, can contaminate the work or storage area, and can become airborne if disturbed. In some cases, depending on the research done in a lab, this dust can also be radioactive.



Fig. 1 the top lead brick exhibits no oxidized surface dust. The bottom lead brick exhibits oxidized surface dust.

Lead dust can also be created when lead is mechanically disturbed due to sanding, cutting, grinding, or filing. This dust can be transferred to hands or clothing and can accumulate on floors and other laboratory surfaces.

Metallic lead itself can be toxic if ingested or inhaled and exposure can cause potential acute or chronic [health effects](#). Acute, short term health effects may include headaches, insomnia, nausea, fatigue, miscarriage, lower IQ in children, etc. Chronic, long term health effects may include anemia, infertility, hypertension, kidney damage, fine tremors, etc. It must be noted that “lead poisoning” may occur from both acute and chronic exposures.

To prevent exposure to lead, follow the hierarchy of controls:

Engineering Controls

1. Purchase and use pre-wrapped/encapsulated lead bricks or weights to help reduce contamination.
2. Encapsulation materials such as epoxy, vinyl, plastisol, polyurethane, thermosetting acrylics, powder coating, and latex paint can be used to coat lead bricks used for radiation shielding to minimize exposure to lead. Paint lead using a paint brush to prevent aerosolization of any lead dust. The encapsulant should be compatible with the type of research done in a lab in order to avoid activation of the coating material.
3. Wrap lead bricks using materials such as certain plastics sheeting, aluminum sheeting, or contact paper. Wear appropriate PPE (see Personal Protective Equipment) when wrapping or painting lead bricks.

4. Choose interlocking lead bricks over flat lead bricks. The interlocking design ensures that there will be no radiation leaks between the bricks.

Administrative Controls

1. Wherever possible, avoid purchasing new metallic lead, and reuse existing metallic lead, which might be available on-site, or transferred from another laboratory or department. If new lead shielding is needed, purchase it wrapped or coated.
2. Avoid machining, drilling, sanding, and cutting metallic lead. Contact EH&S at 650-723-0448, for assessment if you plan to cut lead shielding, since this activity can generate high levels of lead dust.
3. Wash hands and any exposed skin surfaces with soap and water after handling lead bricks.
4. Lead bricks must not be used as counterweights or as door stops.
5. If your work area needs a large amount of lead shielding for radiation protection, make sure the floor and any furniture involved can handle the weight. Ensure the shielding design addresses structural and non-structural support issues related to seismic risks.
6. Lead used for radiation shielding from radioactive sources must be surveyed for radiation prior to transport, reuse, or disposal. Contact EH&S at 650-723-0448 to schedule a radiation survey of lead bricks.
7. Regularly clean work areas around lead shielding, using wet cleaning methods (see Cleaning). The material used for cleaning must be handled and disposed of as hazardous waste.
8. Avoid putting labels or stickers on lead, as they can make future reuse or recycling very difficult.
9. Never throw lead or lead dust in regular trash, or in radioactive or hazardous waste containers. Contact EH&S at 650-723-0448 for recycling or disposal.

Personal Protective Equipment (PPE)

1. When handling uncovered lead bricks, wear disposable gloves, a disposable lab coat or Tyvek coveralls, and safety glasses. This practice will help reduce contamination of clothing. Dispose of used gloves, lab coat or Tyvek coveralls as hazardous lead waste.
2. Respiratory protection may be required if the work involves prolonged duration of stacking or handling oxidized lead bricks or involves a work activity that may agitate lead, such as cutting or grinding, thereby producing lead dust, especially if appropriate engineering controls (e.g., chemical fume hoods, local exhaust ventilation system) are not in place. Contact EH&S at 650-723-0448 for an assessment.

Storage of Lead

1. Lead bricks should be stored in a safe place and in a manner that prevents personnel from having direct contact with metallic lead.
2. Never leave lead lying around unless a lab is actively using it. Store lead pieces in a closed box with appropriate labeling, e.g., "*Lead Shielding for Re-Use*".
3. Do not store lead shielding outdoors even when wrapped in plastic. Condensation may occur, promoting surface oxidation of lead.
4. Metallic lead must be stored in areas that minimize oxidation and degradation of lead. Lead oxides on the surface are more likely to become airborne and contaminate other surfaces, leading to potential exposures.

5. Store metallic lead indoors or in a container that is cool, dry, well ventilated, out of direct sunlight, and away from chemicals, heat, and ignition sources.
6. If it is not feasible to place the bricks in a container, then ensure they are placed on plastic sheeting or rubber mats in a designated area away from traffic corridors or frequently used equipment. Replace the sheeting or mat periodically to prevent accumulation of lead dust.
7. Uncoated lead bricks or weights, or those with deteriorating coatings, must be stored appropriately, so that surrounding surfaces and equipment are not contaminated with lead dust.
8. When storing large quantities of lead bricks and/or weights, address structural and non-structural support issues related to seismic risks. Pay close attention to the consolidated weight of the bricks and the height of the stack of the lead bricks.

Transferring Lead Bricks

1. When transferring metallic lead from one area or department to another, bag the materials in plastic bags or place them in a sturdy lidded container and move in a manner that minimizes the potential for personal exposure and environmental contamination. Protect from contact with rain to prevent release of lead.
2. Small quantities of oxidized lead can still be transported by the user, but extra care must be taken. PPE must always be worn when handling oxidized lead. If feasible, place the oxidized lead bricks under a fume hood, and then put them in a sturdy container prior to moving. The area where the lead was held must also be thoroughly cleaned using wet methods (see Cleaning) and vacuum with a certified HEPA filter.
3. See Ergonomics for proper lifting techniques.

Cleaning

1. Use wet cleaning methods (e.g., water and detergent, misting sprayer, wet cloth/mop, sponges, [lead-safe dust wipes](#), etc.) to clean surfaces, capture debris, and to contain lead dust.
2. [Qualitative surface sampling](#) (i.e., the wipes change color in presence of lead) can be performed to verify cleaning protocols or identify where additional cleaning is required.
3. For larger areas contaminated with lead dust, vacuum with a certified HEPA filter and have it designated specifically for lead cleanup.
4. Perform vacuum maintenance wearing appropriate PPE and dispose of lead dust as hazardous waste.
5. Do not dry-sweep lead dust. Use a wet mop or a vacuum with a certified HEPA filter.

Ergonomics

1. Lead bricks are generally very heavy. Use correct lifting techniques when handling and transporting lead bricks and weights, to avoid ergonomic injuries such as back sprains, muscle pulls, and wrist, elbow, & spinal injuries. See [EH&S-1400](#) “Back Care/Safe Lifting and Carrying” in STARS.
2. Lift only one brick at a time using both hands and avoid reaching or twisting while lifting. Set bricks down gently to minimize dust generation and beware of fingers and toes.
3. Lift no more than one 27-pound brick at a time. Limit lifting to no more than 50 pounds.
4. Use mechanical lifting equipment such as carts, forklifts with pallets, and other methods if feasible, when working with lifting multiple lead bricks.

5. Make sure that the lifting equipment and means of transport are clear of lead contamination after use. Only use equipment such as forklifts if you have been trained and approved to use this equipment.

References

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